Collision divisors and the Laurent locus

The jump criterion for Motegi's type C structure constants, proved;
and the determinant/Pfaffian stratification shown not to be it

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Abstract

Two results, one negative and one positive, about the same object: the structure constants $c^{\nu}_{\lambda\mu}$ of the fixed-point algebra of Motegi's type C trigonometric weight functions (arXiv:2610.15344).

(1) Q492 asked whether the determinant/Pfaffian dichotomy for type B/C boundary partition functions is the *same* stratification as the Laurent criterion $|\nu| - \max(|\lambda|, |\mu|) \leq 1$ measured in Q484. It is not, and the reason is structural rather than numerical. The integer governing Motegi's boundary degeneration is $n - s$, not $n - r$; the number n is a gauge parameter annihilated by his own stability relation; and the gauge-invariant residue $s - r = |\mathbf{M}| - |\mathbf{I}|$ is a cardinality gap between an *evaluation point* and a *basis label*, whereas the jump is a gap among *three basis labels*. The two integers are gaps in different slots of the same matrix. Q492 is closed as a negative (Theorem 1).

(2) The criterion itself, which Q484 could only measure (225 triples, zero error in either direction), is *proved* in one direction, and the mechanism is identified. The candidate pole divisors of $c^{\nu}_{\lambda\mu}$ split into *collision divisors* — loci on which two of the 2^N specialisation points literally collide — and the *long-root divisors* $w_a w_b = 1$, which are not collisions. On a collision divisor a row collapse forces the numerator of the triangular recursion to vanish to matching order, so no pole can occur, for *any* (λ, μ, ν) (Theorem 14). On the long-root divisors a pole is created exactly when a certain two-element deletion stays above both λ and μ, and that condition is equivalent — by a two-line cardinality argument — to jump ≥ 2 (Lemma 16). Hence jump ≤ 1 implies Laurent (Theorem 18). The converse requires a non-cancellation statement which is measured, not proved, and is stated as Gap 2.

The mechanism predicts more than the criterion: it predicts the *exact pole divisor* of each structure constant, hyperplane by hyperplane. Verified 20/20 at $N = 2$ and 115/115 at $N = 3$ against SymPy, by an independent combinatorial mechanism; both pre-registered plants fire.

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1 The gate: Q492 is a one-coordinate match

The session brief armed a thirty-minute gate before any machinery, because two integers were in play and they shared a noun. The gate ran, and it closed the question.

1.1 What Motegi's integer actually is

Fix N and let $\mathbf{w} = (w_1, \ldots, w_N)$. Motegi's partition functions $F_{N,n,r}(\mathbf{u}|\mathbf{w}|\mathbf{I})$, and their type C and B specialisations $WC_{N,n,r}$, $WB_{N,n,r}$, carry three integers (L331–L470 of the e-print source `typebcweightfunctions_arxivversionfinal-3.tex`):

- N, the number of columns, carrying $\mathbf{w}$;
- n, the number of rows, carrying $\mathbf{u} = (u_1, \ldots, u_n)$;
- $r = |\mathbf{I}|$, the number of 1's on the bottom boundary, where $\mathbf{I} = (i_1 < \cdots < i_r)$ equivalently encodes a strict partition $N \geq \lambda_1 > \cdots > \lambda_r \geq 1$ by $i_j = \lambda_{r+1-j}$.

The specialisation is indexed by a *fourth* datum $\mathbf{M} = \{m_1 < \cdots < m_s\} \subseteq [N]$, with

$$\mathbf{u} = \mathbf{w}_{\mathbf{M}} := (w_{m_1}, \ldots, w_{m_s}, \underbrace{1, \ldots, 1}_{n-s}). \tag{1}$$

Motegi states the distinction himself, at L2790–L2803:

> "Thus r specifies the label cardinality, whereas s specifies the number of selected w-variables in the specialization."

So the number of spectral parameters set to 1 is $n - s$, **not** $n - r$.

My own wake note had recorded $n - r$, quoting L2094: "upon setting $u_{r+1} = \cdots = u_n = 1$, the configuration in the top $(n - r)$ rows is uniquely determined by the condition $[K(1)]_0^1 = 0$." That quotation is accurate and the reading of it was wrong: the proposition it belongs to is the *equal-cardinality* case $s = r$, where $\mathbf{w}_{\mathbf{M}} = (w_{m_1}, \ldots, w_{m_r}, 1, \ldots, 1)$, so that $n - s$ and $n - r$ coincide. The quantity was read off a stratum on which the two integers agree and then carried off it.

1.2 And n is gauge

Worse for the hypothesis: n is not an invariant of anything. Motegi's stability relation (`stabilityWC`, L2641–L2647) reads

$$\widetilde{WC}_{N,n,r}(\mathbf{u}|\mathbf{w}|\mathbf{I})\big|_{u_n=1} = \widetilde{WC}_{N,n-1,r}(\mathbf{u}\setminus\{u_n\}|\mathbf{w}|\mathbf{I}), \qquad r \le n-1, \tag{2}$$

and iterating it (his `eq:typeC-stability-s-ge-r`, L2823–L2830) gives, for $s \ge r$,

$$\widetilde{WC}_{N,n,r}(\mathbf{w_M}|\mathbf{w}|\mathbf{I}) = \widetilde{WC}_{N,s,r}(w_{m_1},\dots,w_{m_s}|\mathbf{w}|\mathbf{I}). \tag{3}$$

The normalised weight function — and therefore every structure constant built from it — is *independent of* n once $n \ge s$. An integer containing a free gauge parameter cannot equal a function of (λ,μ,ν): on Q484's smallest failure witness $\lambda=\mu=\emptyset$, $\nu=\{1,2\}$ at $N=2$, the quantity $n-r$ takes *every* value ≥ 0 as n ranges over $n \ge 2$, including 2 and including 0.

Theorem 1 (the gate, Q492 closed negatively)**.** *The determinant/Pfaffian stratification parameter of Motegi's boundary specialisation is not the Q484 jump. Precisely:*

(i) the number of spectral parameters specialised to 1 *in* (1) *is* $n-s$*, and* n *is a gauge parameter removed by* (3)*, so the only gauge-invariant content of that count is* $s-r = |\mathbf{M}|-|\mathbf{I}|$*;*

(ii) $|\mathbf{M}|-|\mathbf{I}|$ *is a cardinality gap between an* evaluation point $\mathbf{M}$ *and a* basis label $\mathbf{I}$ *— a (row, column) pair of the transition matrix* $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ *of §2 — whereas* $|\nu|-\max(|\lambda|,|\mu|)$ *is a gap among* three basis labels*, i.e. three columns;*

(iii) consequently the two integers are cardinality gaps in different slots of the same matrix, and the agreement noted in the wake note (both are "cardinality gaps between increasing sequences in $[N]$*", both vanish on the diagonal* $\mathbf{M}=\mathbf{I}$*) is a match on one coordinate of a two-coordinate object.*

Remark 2. This is the fourth instance this month of one error shape: a quantity that matches on a name and on a single value, carried between two slots. It cost thirty minutes here because the gate was pre-registered and run at position zero. The distinction in (ii) is not a technicality — it is what makes the rest of this paper possible, because the *correct* two-slot statement, the one about rows, turns out to be the engine (§5).

Remark 3 (a false collision, recorded not re-litigated). Motegi's paper says "Laurent" 22 times and proves Laurent polynomiality (L152–153, §L3276). That theorem is about *his specialisations* $\widetilde{WC}_{N,n,r}(\mathbf{w_M}|\mathbf{w}|\mathbf{I})$; Q484's criterion is about the *structure constants* $c^{\nu}_{\lambda\mu}$ in that basis. Same noun, different object: his theorem neither implies nor contradicts mine. The correct relation between them, found in the course of this proof, is that his theorem is an *input* to mine — it is exactly the regularity statement Lemma 7 needs. Neither support nor refutation: a hypothesis.

2 Setup: the fixed-point algebra

We work in type C throughout; $K^C(u) = K(u)|_{\gamma=0,\beta=1}$ (`triandc`, L374). Let $\mathbb{F} := \mathbb{C}(q)(w_1,\dots,w_N)$.

Definition 4. For $\mathbf{M},\mathbf{I} \subseteq [N]$ set $\widetilde{A}_{\mathbf{M},\mathbf{I}} := \widetilde{WC}_{N,n,|\mathbf{I}|}(\mathbf{w_M}|\mathbf{w}|\mathbf{I})$ for any $n \ge \max(|\mathbf{M}|,|\mathbf{I}|)$. This is well defined — independent of n — by (3) for $|\mathbf{M}| \ge |\mathbf{I}|$ and by `eq:typeC-stability-s-lt-r` (L2836–L2840) together with $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ for $|\mathbf{M}| < |\mathbf{I}|$ (L2843–L2848) otherwise. We call $n = |\mathbf{M}|$ the *unpadded* and $n > |\mathbf{M}|$ a *padded* convention, and we use whichever is convenient.

Let V be the algebra of $\mathbb{F}$-valued functions on the 2^N specialisation points, with pointwise product, and let $\varphi_{\mathbf{I}} \in V$ be $\mathbf{M} \mapsto \widetilde{A}_{\mathbf{M},\mathbf{I}}$. Q484 (Lemma 2) proves $\{\varphi_{\mathbf{I}}\}$ is a basis of V, and we write

$$\varphi_\lambda \varphi_\mu = \sum_\nu c^\nu_{\lambda\mu} \varphi_\nu, \qquad \text{equivalently} \qquad \widetilde{A}_{\mathbf{M},\lambda} \widetilde{A}_{\mathbf{M},\mu} = \sum_\nu c^\nu_{\lambda\mu} \widetilde{A}_{\mathbf{M},\nu} \ \ \forall \mathbf{M}. \tag{4}$$

Throughout, λ, μ, ν denote subsets of $[N]$ and $|\cdot|$ denotes *cardinality* (Motegi's own convention: "$r = |\mathbf{I}|$ is the cardinality of the label $\mathbf{I}$", L2790).

We use three facts about $\widetilde{A}$, all from Motegi.

Lemma 5 (symmetry)**.** *$\widetilde{A}_{\mathbf{M},\mathbf{I}}$ depends on $\mathbf{w}_{\mathbf{M}}$ only through the multiset of its entries.*

Proof. $F_{N,n,r}(\mathbf{u}|\mathbf{w}|\mathbf{I})$ is symmetric in $\mathbf{u}$ (L457–L460, by the Yang–Baxter and reflection equations); the normalisation $\mathcal{E}^C_{N,n,r}$ (L1057–L1062) is symmetric in $\mathbf{u}$ as well. □

Lemma 6 (triangularity)**.** *$\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ unless $\mathbf{M} \geq \mathbf{I}$, where Motegi's order is $\mathbf{M} \geq \mathbf{I} \iff |\mathbf{M}| \geq |\mathbf{I}|$ and $m_{s-r+j} \geq i_j$ for $1 \leq j \leq r$.*

Proof. Theorem 2.6(i) of the source (`ffunctionproperties`), L625–L636; restated as `eq:typeC-shifted-dominan` L3250–L3258. (Q484 measured the converse as well, 84/84 pairs at $N \leq 3$ with no violation in either direction; the converse is *not* used below except where flagged.) □

Lemma 7 (regularity)**.** *$\widetilde{A}_{\mathbf{M},\mathbf{I}} \in \mathbb{C}(q)[w_1^{\pm 1}, \ldots, w_N^{\pm 1}]$. Hence $\operatorname{ord}_D \widetilde{A}_{\mathbf{M},\mathbf{I}} \geq 0$ for every prime divisor D other than the coordinate divisors $w_a = 0$.*

Proof. Motegi, "Laurent polynomiality" for the specialisations: claimed L152–153 ("We prove support conditions, Laurent polynomiality, divisibility, and Newton degree bounds for these specializations"), proved §L3276ff. by equivariant geometric localisation with λ_y-classes. Deep-read firsthand. □

Fix a linear extension $\prec$ of Motegi's order. By Lemmas 6 and 7, (4) is a triangular system, solved by forward substitution:

$$\widetilde{A}_{\nu,\nu}\, c^\nu_{\lambda\mu} = \widetilde{A}_{\nu,\lambda} \widetilde{A}_{\nu,\mu} - \sum_{\kappa \prec \nu} c^\kappa_{\lambda\mu} \widetilde{A}_{\nu,\kappa}. \tag{5}$$

In particular $\det(\widetilde{A}_{\mathbf{M},\mathbf{I}}) = \pm \prod_{\mathbf{I}} \widetilde{A}_{\mathbf{I},\mathbf{I}}$, so *every pole of every $c^\nu_{\lambda\mu}$ lies on the vanishing locus of some diagonal entry.*

3 The four candidate divisors

Motegi evaluates the diagonal in closed form (`normalizationWC`, L1078, written out via L1017–L1022 and L1041–L1046): with $\mathbf{I}^{\mathrm{c}} = [N] \setminus \mathbf{I}$,

$$\widetilde{A}_{\mathbf{I},\mathbf{I}} = \underbrace{\prod_{1 \leq j \leq k \leq r} \left(1 - w_{i_j}^{-1} w_{i_k}^{-1}\right)}_{\text{(L) and (S)}} \cdot \underbrace{\prod_{j \in \mathbf{I}^{\mathrm{c}}} \prod_{i_l > j} \left(1 - w_j w_{i_l}^{-1}\right)}_{\text{(A)}} \cdot \underbrace{\prod_{\substack{j \leq k \\ j,k \in \mathbf{I}^{\mathrm{c}}}} \left(1 - q w_j w_k\right) \prod_{j \in \mathbf{I}^{\mathrm{c}}} \prod_{i_l < j} \left(1 - q w_j w_{i_l}^{-1}\right)}_{\text{(Q)}}. \tag{6}$$

This is a product over the positive roots of C_N, and it partitions the candidate pole divisors into exactly four families:

family	divisor	from	disposed of by
(L) long root	$w_a w_b = 1$, $a < b$	$j < k$ in (6)	§6 (jump-governed)
(S) short root	$w_a = \pm 1$	$j = k$ in (6)	§5 (collision)
(A) type A root	$w_a = w_b$, $a < b$	(A) in (6)	§5 (collision)
(Q) q-shifted	$q w_a w_b = 1$, $q w_a = w_b$	(Q) in (6)	§7 (Gap 1)

Naming which control covers which block is the point of the table: three of the four families are settled unconditionally, the first is the content of the theorem, and the last is the one honest gap.

4 Orders of vanishing, from Motegi's branching formula

Everything in §6 rests on one lemma, and Motegi's cross-sector branching formula gives it outright. For $\mathbf{K} \subseteq \mathbf{M}$ with $|\mathbf{K}| = r$, put $\mathbf{L} := \mathbf{M} \backslash \mathbf{K} = (l_1 < \cdots < l_{s-r})$. His `eq:typeC-one-branch-explicit` (L3203–L3236) expresses $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ as $\sum_{\mathbf{K}} T_{\mathbf{K}}$ with

$$
\begin{aligned}
T_{\mathbf{K}} = (1-q)^{s-r} \prod_{\substack{j \le k \\ j,k \in \mathbf{M}^{\mathrm{c}}}} (1 - q w_j w_k) \cdot \frac{W_{N,r}(w_{k_1}, \ldots, w_{k_r} | \mathbf{w} | \mathbf{I})}{\prod_{a,b=1}^{r} (1 - q w_{k_a}^{-1} w_{k_b})} \\
\times \frac{\displaystyle\prod_{1 \le a \le b \le r} \left(1 - w_{k_a}^{-1} w_{k_b}^{-1}\right) \prod_{a=1}^{r} \prod_{b=1}^{s-r} \left(1 - w_{k_a}^{-1} w_{l_b}^{-1}\right)}{\displaystyle\prod_{a=1}^{r} \prod_{b=1}^{s-r} \left(1 - w_{k_a}^{-1} w_{l_b}\right)} \\
\times \prod_{1 \le a < b \le s-r} \left(1 - q w_{l_a}^{-1} w_{l_b}^{-1}\right) \prod_{a=1}^{s-r} \prod_{j \in \mathbf{M}^{\mathrm{c}}} \left(1 - q w_{l_a}^{-1} w_j\right). \qquad (7)
\end{aligned}
$$

Read off which pairs $\{x, y\} \subseteq \mathbf{M}$ contribute a factor vanishing on the *long-root* divisor $w_x w_y = 1$:

- both $x, y \in \mathbf{K}$: yes, from $\prod_{a<b\le r}(1 - w_{k_a}^{-1} w_{k_b}^{-1})$;
- exactly one of x, y in $\mathbf{K}$: yes, from $\prod_a \prod_b (1 - w_{k_a}^{-1} w_{l_b}^{-1})$;
- neither in $\mathbf{K}$ (both in $\mathbf{L}$): **no** — such pairs appear only as $\left(1 - q w_{l_a}^{-1} w_{l_b}^{-1}\right)$, which vanishes on the q-shifted divisor $q w_x w_y = 1$, a different hyperplane.

No denominator in (7) vanishes on a long-root divisor: the denominators are $\prod(1 - q w_{k_a}^{-1} w_{k_b})$ and $\prod(1 - w_{k_a}^{-1} w_{l_b})$, of families (Q) and (A). Hence:

Lemma 8 (the branch-term order)**.** *Let $a < b$ with $\{a, b\} \subseteq \mathbf{M}$. Then $\operatorname{ord}_{\{w_a w_b = 1\}} T_{\mathbf{K}} \ge 1$ whenever $\mathbf{K} \cap \{a, b\} \ne \emptyset$.*

Which $\mathbf{K}$ survive is also Motegi's:

Lemma 9 (admissibility)**.** *$T_{\mathbf{K}} = 0$ unless $\mathbf{K} \ge \mathbf{I}$, i.e. $k_j \ge i_j$ for all j. Moreover, for any $P \subseteq [N]$, there exists $\mathbf{K} \subseteq P$ with $|\mathbf{K}| = r$ and $\mathbf{K} \ge \mathbf{I}$ if and only if $P \ge \mathbf{I}$ in Motegi's order.*

Proof. The first claim is type A triangularity applied to the factor $W_{N,r}(w_{\mathbf{K}} | \mathbf{w} | \mathbf{I}) / \prod_{a,b}(1 - q w_{k_a}^{-1} w_{k_b})$: Motegi, L3263–L3269. The second is his L3269–L3272 verbatim — "the necessary and sufficient condition for $\mathbf{M}$ to contain an r-element subset $\mathbf{K} \ge \mathbf{I}$ is $\{m_{s-r+1}, \ldots, m_s\} \ge \mathbf{I}$" — which is the definition of $P \ge \mathbf{I}$, since that order compares exactly the top r elements of P with $\mathbf{I}$. □

Lemma 10 (Lemma A: the long-root order)**.** *Let $a < b$ and $H_{ab} := \{w_a w_b = 1\}$. If $\{a, b\} \subseteq \mathbf{M}$ and $\mathbf{M} \setminus \{a, b\} \not\geq \mathbf{I}$, then* $\operatorname{ord}_{H_{ab}} \widetilde{A}_{\mathbf{M},\mathbf{I}} \geq 1$.

Proof. By Lemma 9, $\widetilde{A}_{\mathbf{M},\mathbf{I}} = \sum_{\mathbf{K} \geq \mathbf{I}} T_{\mathbf{K}}$. Suppose some surviving $\mathbf{K}$ had $\mathbf{K} \cap \{a, b\} = \emptyset$; then $\mathbf{K} \subseteq \mathbf{M} \setminus \{a, b\}$ with $|\mathbf{K}| = r$ and $\mathbf{K} \geq \mathbf{I}$, so by Lemma 9 (second part, with $P = \mathbf{M} \setminus \{a, b\}$) we would get $\mathbf{M} \setminus \{a, b\} \geq \mathbf{I}$, contrary to hypothesis. So every surviving $\mathbf{K}$ meets $\{a, b\}$, and Lemma 8 makes every surviving term divisible by $(1 - w_a^{-1} w_b^{-1})$. □

Remark 11. The same reading of (7) at $a = b$ gives the (S) analogue: $\operatorname{ord}_{\{w_a = \pm 1\}} \widetilde{A}_{\mathbf{M},\mathbf{I}} \geq 1$ if $a \in \mathbf{M}$ and $\mathbf{M} \setminus \{a\} \not\geq \mathbf{I}$, from the $a = b$ term $(1 - w_{k_a}^{-2})$. We will not need it — §5 proves more — but it is the reason the measured order table (§8) has the same shape in both families.

5 Collision divisors carry no pole, ever

This section is unconditional: no hypothesis on λ, μ, ν.

Definition 12. A prime divisor D is a *collision divisor* with *collapse map* τ if there is a set $\mathcal{D}$ of subsets of $[N]$ and a map $\tau : \mathcal{D} \to 2^{[N]}$ such that for every $\mathbf{M} \in \mathcal{D}$:

$$\tau(\mathbf{M}) < \mathbf{M} \quad \text{and} \quad \widetilde{A}_{\mathbf{M},\mathbf{I}}\big|_D = \widetilde{A}_{\tau(\mathbf{M}),\mathbf{I}}\big|_D \ \text{ for all } \mathbf{I}, \tag{8}$$

and such that $\operatorname{ord}_D \widetilde{A}_{\mathbf{M},\mathbf{M}} = 0$ for $\mathbf{M} \notin \mathcal{D}$.

Lemma 13 (the collapses)**.** *(i) $D = \{w_a = 1\}$ is a collision divisor with $\mathcal{D} = \{\mathbf{M} : a \in \mathbf{M}\}$, $\tau(\mathbf{M}) = \mathbf{M} \setminus \{a\}$.*

(ii) $D = \{w_a = w_b\}$, $a < b$, is a collision divisor with $\mathcal{D} = \{\mathbf{M} : b \in \mathbf{M},\, a \notin \mathbf{M}\}$, $\tau(\mathbf{M}) = (\mathbf{M} \setminus \{b\}) \cup \{a\}$.

(iii) $\widetilde{A}_{\mathbf{M},\mathbf{I}}(-\mathbf{w}) = \widetilde{A}_{\mathbf{M},\mathbf{I}}(\mathbf{w})$; consequently $D = \{w_a = -1\}$ is a collision divisor with the same data as in (i).

Proof. (i) Work in a padded convention with n fixed, $n \geq |\mathbf{M}|$. On $\{w_a = 1\}$ the tuple $\mathbf{w}_{\mathbf{M}}$ has entries $\{w_m : m \in \mathbf{M} \setminus \{a\}\}$ together with $n - |\mathbf{M}| + 1$ copies of 1; the tuple $\mathbf{w}_{\mathbf{M} \setminus \{a\}}$, padded to the same n, has the same multiset. Lemma 5 gives (8). For $\tau(\mathbf{M}) < \mathbf{M}$: write $a = \nu_i$ in the increasing listing of $\mathbf{M}$; then $|\mathbf{M}| = |\tau\mathbf{M}| + 1$ and the top $|\tau\mathbf{M}|$ entries of $\mathbf{M}$ dominate $\tau\mathbf{M}$ entrywise, so $\mathbf{M} \geq \tau\mathbf{M}$, and $\neq$.

(ii) On $\{w_a = w_b\}$ with $b \in \mathbf{M}$, $a \notin \mathbf{M}$, the multisets of entries of $\mathbf{w}_{\mathbf{M}}$ and $\mathbf{w}_{(\mathbf{M} \setminus \{b\}) \cup \{a\}}$ agree; again Lemma 5. Both sets have the same cardinality and $\tau\mathbf{M}$ is obtained from $\mathbf{M}$ by decreasing one element, so $\mathbf{M} > \tau\mathbf{M}$.

(iii) Use the unpadded convention $n = |\mathbf{M}| = s$ (legitimate, the definition is n-independent; for $s < r$ both sides vanish). Then $(\mathbf{u}, \mathbf{w}) \mapsto (-\mathbf{u}, -\mathbf{w})$, and we count sign changes in the bracket of $F_{N,n,r}$ at $\beta = 1$, $\gamma = 0$ (L474–L487). Invariant: $(u_j^2 - 1)$, $(u_j u_k - 1)$, $(u_j u_k - q)$, and each ratio $(qu_j - u_k)/(u_j - u_k)$. Sign-changing: $(u_j - q w_k)$ for $j \leq r$, $k = i_j + 1, \ldots, N$, contributing $\sum_j (N - i_j)$; $(u_j - w_k)$ for $j \leq r$, $k < i_j$, contributing $\sum_j (i_j - 1)$; $(u_j - q w_k)$ for $j > r$, all k, contributing $(n - r)N$; and $(\beta u_j - \gamma) = u_j$ for $j > r$, contributing $(n - r)$. The prefactor $\prod_k u_k^{-(N+n)}$ contributes $n(N + n)$. Total

$$\textstyle\sum_j (N - i_j) + \sum_j (i_j - 1) + (n - r)N + (n - r) + n(N + n) = 2nN - 2r + n + n^2 \equiv n(n+1) \equiv 0 \pmod 2.$$

Every factor of $\mathcal{E}^C_{N,n,r}$ — $(1-qu_k/u_j)$, $(1-qu_jw_k)$, $(1-qu_ju_k)$, $(1-qw_jw_k)$ — is invariant. So $\widetilde{A}$ is invariant. Now $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_{w_a=-1}$ equals $\widetilde{A}_{\mathbf{M},\mathbf{I}}(\mathbf{v})|_{v_a=1}$ under $\mathbf{w}=-\mathbf{v}$, which by (i) equals $\widetilde{A}_{\mathbf{M}\setminus\{a\},\mathbf{I}}(\mathbf{v})|_{v_a=1} = \widetilde{A}_{\mathbf{M}\setminus\{a\},\mathbf{I}}|_{w_a=-1}$. □

Theorem 14 (collision divisors)**.** *Let D be a collision divisor. Then $\operatorname{ord}_D c^\nu_{\lambda\mu} \geq 0$ for all λ,μ,ν. In particular no $c^\nu_{\lambda\mu}$ has a pole on $w_a=1$, on $w_a=-1$, or on $w_a=w_b$.*

Proof. Induction along $\prec$. Assume $\operatorname{ord}_D c^\kappa_{\lambda\mu} \geq 0$ for all $\kappa \prec \nu$ (vacuous at the start). By Lemma 7 every $\widetilde{A}$ is regular on D.

If $\nu \notin \mathcal{D}$ then $\operatorname{ord}_D \widetilde{A}_{\nu,\nu} = 0$ and (5) gives $\operatorname{ord}_D c^\nu \geq 0$ immediately.

So let $\nu \in \mathcal{D}$ and put $\nu' := \tau(\nu) < \nu$. Since $|\nu'| < |\nu|$ or $\nu' \neq \nu$ with $\nu > \nu'$, triangularity gives $\widetilde{A}_{\nu',\nu} = 0$, hence by (8) $\operatorname{ord}_D \widetilde{A}_{\nu,\nu} \geq 1$. We show the right side of (5) vanishes on D to at least the same order; it suffices to show it vanishes on D, since $\operatorname{ord}_D \widetilde{A}_{\nu,\nu}$ will be matched after dividing — precisely, we show

$$\Big[\widetilde{A}_{\nu,\lambda}\widetilde{A}_{\nu,\mu} - \sum_{\kappa\prec\nu} c^\kappa \widetilde{A}_{\nu,\kappa}\Big]\Big|_D = 0. \tag{9}$$

Apply (8) to each $\widetilde{A}_{\nu,\bullet}$: the left side of (9) equals $\big[\widetilde{A}_{\nu',\lambda}\widetilde{A}_{\nu',\mu} - \sum_{\kappa\prec\nu} c^\kappa \widetilde{A}_{\nu',\kappa}\big]|_D$, all terms being regular on D by Lemma 7 and the inductive hypothesis. Now evaluate (4) at $\mathbf{M}=\nu'$:

$$\widetilde{A}_{\nu',\lambda}\widetilde{A}_{\nu',\mu} = \sum_\kappa c^\kappa \widetilde{A}_{\nu',\kappa},$$

where by Lemma 6 the sum is over κ with $\nu' \geq \kappa$. Every such κ satisfies $\kappa \leq \nu' < \nu$, hence $\kappa \prec \nu$; and every $\kappa \prec \nu$ with $\nu' \not\geq \kappa$ has $\widetilde{A}_{\nu',\kappa} = 0$. So the two sums agree term by term, and (9) holds.

Finally, (9) says the numerator lies in the prime ideal of D, so $\operatorname{ord}_D(\text{numerator}) \geq 1$. It remains to check $\operatorname{ord}_D \widetilde{A}_{\nu,\nu} = 1$, i.e. that the collision divisor occurs in the diagonal entry to multiplicity exactly one. This is visible in (6): for $D=\{w_a=\varepsilon\}$ the only factor of $\widetilde{A}_{\nu,\nu}$ vanishing on D is the single $j=k=a$ term $(1-w_a^{-2}) = -w_a^{-2}(w_a-1)(w_a+1)$, in which $(w_a-\varepsilon)$ occurs once; for $D=\{w_a=w_b\}$ it is the single factor $(1-w_aw_b^{-1})$ with $a\in\nu^{\mathrm{c}}$, $b\in\nu$, $b>a$. All factors listed in (6) are pairwise non-associate irreducibles, so no multiplicity accumulates. Hence $\operatorname{ord}_D c^\nu \geq 1-1=0$. □

Remark 15 (where this comes from, physically). $[K^C(u)]^1_0 = 1-u^{-2}$ vanishes at $u=\pm1$, which is why a row carrying $u=\pm1$ is forced and removable — that is Motegi's own engine (L1750, L2094), and (2) is its normalised form. The brief asked me to name the component I skip: $1-u^{-2}$ vanishes at *both* $u=1$ and $u=-1$, and Motegi only ever uses $u=+1$. Lemma 13(iii) answers it. The $u=-1$ branch is **not** trivial — it is a genuine second collision divisor, with the same collapse map — and it is reached by the symmetry $\mathbf{w}\mapsto-\mathbf{w}$, not by Motegi's argument. Unlike the $q=-1$ collapse of 2026-10-15, there is mathematics on this component.

6 The long-root divisors: the jump appears

The long-root divisors $w_aw_b=1$ are *not* collision divisors: setting $w_b=w_a^{-1}$ does not make $\mathbf{w}_\mathbf{M}$ coincide with any other specialisation point, because w_a^{-1} is not among the w_c or the padding 1's. This is the whole asymmetry, and it is where poles live.

Lemma 16 (the combinatorial step)**.** *Let $\nu\geq\lambda$ and $\nu\geq\mu$. The following are equivalent:*

(a) there is a pair $a<b$ with $\{a,b\}\subseteq\nu$, $\nu\setminus\{a,b\}\geq\lambda$ and $\nu\setminus\{a,b\}\geq\mu$;

(b) $|\nu| - \max(|\lambda|, |\mu|) \geq 2$.

Proof. (a)⇒(b): $\nu \setminus \{a, b\} \geq \lambda$ forces $|\nu| - 2 \geq |\lambda|$, and likewise $|\nu| - 2 \geq |\mu|$.

(b)⇒(a): let $\{a, b\}$ be the two *smallest* elements of ν (available since $|\nu| \geq 2$), and $\nu' := \nu \setminus \{a, b\}$, so $|\nu'| = |\nu| - 2 \geq \max(|\lambda|, |\mu|)$. Motegi's order compares only the top $|\lambda|$ elements of the larger set with λ; since $|\nu'| \geq |\lambda|$, the top $|\lambda|$ elements of ν' are the top $|\lambda|$ elements of ν. Hence $\nu \geq \lambda$ gives $\nu' \geq \lambda$, and similarly $\nu' \geq \mu$. □

Remark 17. This is where the max in Q484's criterion comes from, and why it looked strange. The criterion reads "$\min(|\nu| - |\lambda|, |\nu| - |\mu|) \leq 1$" — an *or* — and the reason is that a pole needs the deleted pair to stay above *both* λ and μ, an *and* whose negation is an *or*. The two deleted elements are the two smallest of ν; the cost "2" is the size of a long root's support.

Theorem 18 (main: jump ≤ 1 implies no long-root pole)**.** *Suppose* $|\nu| - \max(|\lambda|, |\mu|) \leq 1$. *Then* $\operatorname{ord}_{H_{ab}} c^{\nu}_{\lambda\mu} \geq 0$ *for every* $a < b$. *Combined with Theorem 14 and Gap 1,* $c^{\nu}_{\lambda\mu}$ *is then a Laurent polynomial in* $w_1, \ldots, w_N$ *over* $\mathbb{C}(q)$.

Proof. Write $m := \max(|\lambda|, |\mu|)$ and assume without loss of generality $|\lambda| = m$. Fix $a < b$ and put $H := H_{ab}$. We induct along $\prec$, the claim at ν being $\operatorname{ord}_H c^{\nu} \geq 0$ for all ν with $|\nu| \leq m + 1$.

If $\{a, b\} \not\subseteq \nu$ then $\operatorname{ord}_H \widetilde{A}_{\nu,\nu} = 0$ by (6) — the factor $(1 - w_a^{-1} w_b^{-1})$ occurs there only for $a, b \in \nu$ — and (5) with Lemma 7 and the inductive hypothesis gives $\operatorname{ord}_H c^{\nu} \geq 0$.

So let $\{a, b\} \subseteq \nu$ and $\nu' := \nu \setminus \{a, b\}$. Since $|\nu'| = |\nu| - 2 < |\nu|$, triangularity gives $\nu' \not\geq \nu$, so Lemma 10 applies with $\mathbf{M} = \mathbf{I} = \nu$ and $\operatorname{ord}_H \widetilde{A}_{\nu,\nu} \geq 1$; by (6) the factor occurs to order exactly 1, so $\operatorname{ord}_H \widetilde{A}_{\nu,\nu} = 1$. It therefore suffices to show the right side of (5) has $\operatorname{ord}_H \geq 1$, and we do it termwise.

Term $\widetilde{A}_{\nu,\lambda} \widetilde{A}_{\nu,\mu}$. We have $|\nu'| = |\nu| - 2 \leq m - 1 < m = |\lambda|$, so $\nu' \not\geq \lambda$ for cardinality reasons. Lemma 10 gives $\operatorname{ord}_H \widetilde{A}_{\nu,\lambda} \geq 1$, and $\operatorname{ord}_H \widetilde{A}_{\nu,\mu} \geq 0$ by Lemma 7.

Terms $c^{\kappa} \widetilde{A}_{\nu,\kappa}$, $\kappa \prec \nu$. Only κ with $\widetilde{A}_{\nu,\kappa} \neq 0$ matter, so $\nu \geq \kappa$, hence $|\kappa| \leq |\nu| \leq m + 1$ and the inductive hypothesis gives $\operatorname{ord}_H c^{\kappa} \geq 0$. Only κ with $c^{\kappa} \neq 0$ matter, so by the support bound (Q484, Theorem A) $\kappa \geq \lambda$ and $\kappa \geq \mu$, hence $|\kappa| \geq m$. Then $|\nu'| = |\nu| - 2 \leq m - 1 < m \leq |\kappa|$, so $\nu' \not\geq \kappa$ and Lemma 10 gives $\operatorname{ord}_H \widetilde{A}_{\nu,\kappa} \geq 1$. Therefore $\operatorname{ord}_H (c^{\kappa} \widetilde{A}_{\nu,\kappa}) \geq 0 + 1 = 1$.

Both terms have $\operatorname{ord}_H \geq 1 = \operatorname{ord}_H \widetilde{A}_{\nu,\nu}$, so $\operatorname{ord}_H c^{\nu} \geq 0$. □

Remark 19 (why the induction closes). The step that makes this work is that the hypothesis is inherited: every κ entering the recursion has $|\kappa| \leq |\nu| \leq \max(|\lambda|, |\mu|) + 1$, so κ also has jump ≤ 1 and the inductive hypothesis is available. Equivalently: *jump is monotone in* $|\nu|$ *along Motegi's order*, which is why a pole cannot be created below ν and inherited by a ν of small jump.

6.1 The exact pole divisor: creation and propagation

The mechanism says more than Theorem 18. Reading Lemma 16 locally at each κ rather than at ν gives a prediction of the *divisor*, not just of Laurent-ness.

Conjecture 20 (exact pole divisor; verified, see §8). For all λ, μ, ν the set of long-root divisors on which $c^{\nu}_{\lambda\mu}$ has a pole is

$$\big\{ H_{ab} \;:\; \exists\, \kappa \text{ with } \nu \geq \kappa \geq \lambda,\ \kappa \geq \mu,\ \{a, b\} \subseteq \kappa,\ \kappa \setminus \{a, b\} \geq \lambda,\ \kappa \setminus \{a, b\} \geq \mu \big\}.$$

A pole is *created* at κ when the bracket condition holds there, and then *propagates* upward: for $\nu > \kappa$ the term $c^{\kappa} \widetilde{A}_{\nu,\kappa}$ in (5) carries it, and $\widetilde{A}_{\nu,\kappa}$ need not vanish on H_{ab}. This is why, e.g., $c^{\{1,3\}}_{\emptyset\emptyset}$

at $N = 3$ has a pole on $w_1 w_2 = 1$ although $2 \notin \{1, 3\}$: the pole is created at $\kappa = \{1, 2\} \leq \{1, 3\}$. I had first predicted only the created poles and the measurement corrected me; Conjecture 20 is the corrected statement. Note that it is consistent with Theorem 18: if ν has jump ≤ 1 then every $\kappa \leq \nu$ in the support has $|\kappa| \leq |\nu|$, hence jump ≤ 1, hence by Lemma 16 no κ satisfies the bracket.

7 The q-shifted family

Gap 1 (the q-shifted divisors)**.** *For D in family (Q) — $qw_a w_b = 1$ or $qw_a = w_b$ — I verify computationally (§8) that*

$$\operatorname{ord}_D \widetilde{A}_{\nu,\nu} \leq \operatorname{ord}_D \widetilde{A}_{\nu,\mathbf{I}} \qquad \text{for all } \mathbf{I} \leq \nu, \tag{10}$$

which closes the case at once: with $e := \operatorname{ord}_D \widetilde{A}_{\nu,\nu}$, (5) gives $\operatorname{ord}_D(\text{numerator}) \geq \min(2e, 0 + e) = e$ and so $\operatorname{ord}_D c^{\nu} \geq 0$ by the same induction as before. I do not *prove* (10)*. What I can prove is the structural half: the contribution of the normalisation $\mathcal{E}^C$ to $\operatorname{ord}_D \widetilde{A}_{\mathbf{M},\mathbf{I}}$ depends on $\mathbf{M}$ alone and not on $\mathbf{I}$ — its denominator $\prod_{j \leq k}(1 - qw_j w_k) \prod_{j<k \in \mathbf{M}}(1 - qw_{m_j} w_{m_k})$ and numerator $\prod(1 - qw_{m_k}/w_{m_j}) \prod(1 - qw_{m_j} w_k)$ are functions of $\mathbf{M}$ only — so that* (10) *is a statement about the $\mathbf{I}$-dependence of the $W_{N,r}$ and bottom-boundary blocks of* (7) *alone. Status: verified at $N = 2, 3$ over all $\mathbf{I} \leq \nu$,* 9 *divisors at $N = 2$ and* 9 *at $N = 3$, no violation; not proved.*

Gap 2 (non-vanishing, the converse direction)**.** *Theorem 18 is the "jump $\leq 1 \Rightarrow$ Laurent" half. The converse — jump ≥ 2 implies a pole* actually occurs *— requires that the created pole have nonzero residue, i.e. that the numerator of* (5) *does* not *vanish on H_{ab} when Lemma 16(a) holds. Lemma 10 is a divisibility statement and gives only lower bounds on orders, so it cannot supply this. Equivalently, one needs the converse of Lemma 6 ($\mathbf{M} \geq \mathbf{I} \Rightarrow \widetilde{A}_{\mathbf{M},\mathbf{I}} \neq 0$, measured* 84/84 *in Q484) together with the absence of conspiratorial cancellation among the surviving $\mathbf{K}$-terms of* (7)*. Status: measured.* 225/225 *triples at $N = 2, 3$ in Q484 with zero error in either direction; and, much more sharply, Conjecture 20 holds exactly,* 20/20 *and* 115/115 *(§8).*

8 Computational verification, with controls and plants

All computation is pure Python + SymPy over $\mathbb{C}(q)(w_1, \ldots, w_N)$ (SageMath is not installed in this container). The engine is the Q484 engine `q484_engine.py`, which evaluates $F_{N,n,r}(\mathbf{u}|\mathbf{w}|\mathbf{I})$ by brute-force summation over S_n directly from the symmetrisation formula (L474–L487) — i.e. it does *not* use the branching formula (7) on which §4 rests, so the two are independent. Scripts: `proofs/code-1016-q492/`.

Reproduction of Q484. At $N = 2$, 29/30 triples Laurent, the unique failure $c^{\{1,2\}}_{\emptyset\emptyset}$ with pole divisor $(w_1 w_2 - 1)$. At $N = 3$, 102/115 Laurent on the $\mu \geq \lambda$ half, i.e. 176/195 in full. Jump distribution: jump 0 gives $58 + 12$ Laurent and 0 poles; jump 1 gives $44 + 7$ Laurent and 0 poles; jump 2 gives 0 Laurent and $12 + 1$ poles; jump 3 gives 0 Laurent and 1 pole.

Control 1: the exact pole divisor (independent mechanism). Conjecture 20 is pure combinatorics on subsets; the measured divisor comes from SymPy rational arithmetic and `factor_list`. The two agree on 20/20 nonzero constants at $N = 2$ and 115/115 at $N = 3$. This is a far stronger test than the binary criterion: it checks *which* hyperplanes, with no free parameter. For instance at $N = 3$, $c^{\{1,2,3\}}_{\{1\}\{2\}}$ is predicted to have poles on $w_1 w_2 = 1$ and $w_1 w_3 = 1$ but *not* on $w_2 w_3 = 1$, and the measured denominator is $(w_1 w_2 - 1)(w_1 w_3 - 1)$.

Control 2: no pole on a collision or q-divisor. Theorem 14 predicts that no denominator ever contains a factor from families (S), (A) or (Q). Measured: 20/20 at $N = 2$ and 115/115 at $N = 3$ — every pole divisor observed, across all triples, is a product of long roots $w_a w_b - 1$ only.

Order tables (Lemma 10 and its (S), (A) analogues). Measured as exact equalities, not just the lower bounds the proof uses: $\mathrm{ord}_{H_{ab}} \widetilde{A}_{\mathbf{M},\mathbf{I}} = [\{a,b\} \subseteq \mathbf{M}] \cdot [\mathbf{M} \setminus \{a,b\} \not\supseteq \mathbf{I}]$, confirmed on 10 pairs $(\mathbf{M},\mathbf{I})$ at $N = 2$ and 105 at $N = 3$; $\mathrm{ord}_{\{w_a=\varepsilon\}} \widetilde{A}_{\mathbf{M},\mathbf{I}} = [a \in \mathbf{M}][\mathbf{M} \setminus \{a\} \not\supseteq \mathbf{I}]$ for both $\varepsilon = \pm 1$, confirmed at $N = 2, 3$; $\mathrm{ord}_{\{w_a=w_b\}} \widetilde{A}_{\mathbf{M},\mathbf{I}} = [a \notin \mathbf{M}][b \in \mathbf{M}][(\mathbf{M} \setminus \{b\}) \cup \{a\} \not\supseteq \mathbf{I}]$, confirmed at $N = 2, 3$ — note this last is exactly the collapse map of Lemma 13(ii), measured independently of it. Also verified, since Theorem 14 needs it: on every collision divisor D and for every $\mathbf{I}$, $\mathrm{ord}_D \widetilde{A}_{\mathbf{I},\mathbf{I}} \in \{0, 1\}$ — the multiplicity-one claim at the end of that proof — at $N = 2$ (6 divisors) and $N = 3$ (9 divisors), no value ≥ 2 occurring.

Control 3: the $\mathbf{w} \mapsto -\mathbf{w}$ symmetry. Lemma 13(iii) asserts $\widetilde{A}(-\mathbf{w}) = \widetilde{A}(\mathbf{w})$. Verified on 10/10 nonzero entries at $N = 2$ and 35/35 at $N = 3$. This is the control for the $u = -1$ component, and it is *not* over an inherited domain: the entries are the same ones the order tables run on.

Plant P1 (K-matrix off-diagonal): FIRED. Replace the vanishing factor $(u_j^2 - 1) \mapsto (u_j^2 - q)$ in the $j \leq r$ block, which moves the short-root collision locus from $w_a = \pm 1$ to $w_a^2 = q$ — a locus that is *not* a collision of specialisation points. Theorem 14 should then fail. It does: Control 2 drops to 12/20 and the new denominators are exactly $q - w_a^2$. The plant fires, and it fires *on the predicted family*, which is the informative part.

Plant P2 (R-matrix weight): FIRED. Replace $(u_j u_k - 1) \mapsto (u_j u_k - q)$ in the $1 \leq j < k \leq r$ block, moving the long-root divisor to $q w_a w_b = 1$. Control 1 drops to 19/20 — the $\{1, 2\}$ pole leaves $w_1 w_2 = 1$, exactly as predicted — and Control 2 to 14/20, with new denominators $q - w_a w_b$. Both plants fire.

The stratum each control validates, and what it cannot see. Following the 2026-10-15 fault in which a 14/14 diagonal control was blind to a plant because $\mathbf{M} = \mathbf{I}$ empties the index range $r+1 \leq j < k \leq n$: the controls here run on the *off-diagonal* stratum by construction, since Control 1 and Control 2 are statements about $c^{\nu}_{\lambda\mu}$ for all triples, including $|\nu| > |\lambda|, |\mu|$, where $s > r$ and all blocks of (7) are live. The order tables run over all $(\mathbf{M},\mathbf{I})$ with $\widetilde{A} \neq 0$, which at $N = 3$ is 35 pairs of which 27 are off-diagonal. What the controls do *not* cover: $N \geq 4$ (Lemma 16's content is N-independent, but Conjecture 20 is verified only at $N \leq 3$), and type B (every number here is type C; see §9).

9 Open threads

1. **Gap 2 (the converse).** Proving jump $\geq 2 \Rightarrow$ pole needs the converse of Lemma 6 plus non-cancellation among the $\mathbf{K}$-terms of (7). The natural route is to show the leading coefficient on H_{ab} is a sum of *like-signed* contributions; Motegi's branching formula (7) makes each $T_{\mathbf{K}}$ explicit, so this is a finite computation per $\mathbf{K}$-class and not obviously hard. This is the next prove session.

2. **Gap 1 (the q-family).** Prove (10). Half the statement is reduced to the $\mathbf{I}$-dependence of the $W_{N,r}$ block.

3. **Type B.** Every number in this paper and in Q484 is type C. Nothing in §§5–6 uses $\gamma = 0$ except Lemma 13(iii), whose sign count would have to be redone for WB (there $\gamma = -1$, and the extra prefactor $\prod_k (1 + u_k^{-1})^{-1}$ is *not* invariant under $\mathbf{u} \mapsto -\mathbf{u}$ — it has a pole at $u_k = -1$). So the $u = -1$ collision divisor is exactly where type B may differ from type C, which is a sharper prediction than Q484's open thread 3 and is cheap to test. Motegi defines type B **M**-smallness at L1346 in the same file.

4. $N \geq 4$**.** Conjecture 20 is verified only at $N \leq 3$. Lemma 16 and Theorem 18 are N-independent.

5. **What Q492 was really asking.** Theorem 1 kills the identification of the jump with Motegi's $n - s$. But §5 supplies the structural statement the question wanted, in the slot the gate says is the right one: a puzzle-style rule fails not because the object is "a Pfaffian read in determinant coordinates" but because *the long-root divisors are the ones with no collision to collapse them.* Whether that is related to the determinant/Pfaffian dichotomy of `1806.00933` remains unknown to me: that paper is at extraction level *abstract* and is not on disk. Nothing above depends on it.

Source hygiene

Of the sources this paper composes, exactly one is read first-hand, and it is the one every proof step cites: Motegi, arXiv:2610.15344, e-print source `typebcweightfunctions_arxivversionfinal-3.tex`, 216,486 B, 6430 lines, deep-read. Every line number above was read in that file during this session. The Pfaffian side of the original dichotomy (`1806.00933`, `2309.08446`, `2603.20865`, `math/0008184`) is **unread** — abstract or title-only, none on disk — and consequently *no statement in this paper rests on it*; the word Pfaffian appears here only in §1 and §9 item 5, in both cases to say that the connection is not established. Q484's results (support bound, triangularity converse, the 225-triple measurement) are my own, graded in `proofs/registry/`, and are cited as such.